Course Syllabus

GRAD-E1487: Causal Machine Learning

1. General Information

Course Format	Onsite
Time	Tuesdays, 16:00-18:00
Instructor	Drew Dimmery

Instructor Information

Drew Dimmery is Professor of Data Science for the Common Good at the Hertie School. His research interests include causal inference, machine learning and the usage of data science for public policy. Prior to moving back to academia, he was a research scientist at Facebook in the United States for four years. He holds a PhD in political methodology from New York University and has published in top machine learning journals such as ICML, KDD and AISTATS and general interest journals like Science, Nature and Proceedings of the National Academy of Sciences.

2. Course Contents and Learning Objectives

Course Contents

This course is an introduction and survey of Causal Machine Learning. It examines a wide variety of these approaches and how they can be used in the course of doing causal research in the social sciences. It presumes an existing knowledge of both causal inference and of machine learning. We will read a selection of papers introducing and implementing techniques and research designs, with applications to the social and health sciences and public policy. We will discuss what these new methods are able to offer, and where they may have limitations. The course will be oriented around class discussion and student presentations on the readings. Students will further gain the ability to critically evaluate methodologies through the use of Monte Carlo simulations.

Main Learning Objectives

Students will:

- Learn to use simulations to understand the mechanics of a method/estimator
- Understand how to use machine learning methods for causal questions in the social sciences
- Become proficient in applying these methods to problems

The course is oriented around class discussion and student presentations, so students are expected to take an active role.

Prerequisites

- Has taken Machine Learning
- Has taken Causal Inference
- Comfortable training ML models in Python using sklearn, PyTorch, doing numerical computing using NumPy

Diversity Statement

Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and evaluation of programmes designed for specific groups, populations, or communities. Diversity and cultural awareness will be integrated in the course content whenever possible.

3. Grading and Assignments

Composition of Final Grade

Assignment	Deadline	Submission	Weight
Assignment 1: Referee report	Before corresponding class	Submit via GitHub	15%
Assignment 2: Class presentation	Organized on first day of class	In-class	20%
Assignment 3: Class demonstration	Organized on first day of class	In-class	25%
Assignment 4: Project	End of term (December 11)	Submit via GitHub	30%
Participation grade	-	-	10%

Assignment Details

Assignment 1: Referee Report Select one of the papers on the syllabus — a common, presenter, or optional reading from any session, but *not* a foundational reading. Write a referee report as if you were requested to review the paper for publication at a journal, summarizing the paper's main contributions, areas that could be strengthened, problems and additional extensions or analyses that you would like to see. I expect a concise but information-dense report of 1-2 pages. Submit the referee report on GitHub 36 hours before the paper is discussed in class. Your referee report, class presentation, and class demonstration must each fall in a different week.

References on how to write a referee report:

- Berk, J., C. R. Harvey, and D. A. Hirshleifer (2016). Preparing a referee report: guidelines and perspectives. Available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2547191
- https://chrisblattman.com/blog/2012/01/18/how-to-referee-an-academic-paper/

Preparation of this assignment should take no more than a few hours.

Assignment 2: Class Presentation Each student will be responsible for presenting and leading discussion on one of the **presenter** papers, beginning the third week of the term. Discussion leaders will provide a 10-15 minute high-level overview of the paper presented as a slide-deck, and will develop a list of discussion questions for the class to discuss. For classes with multiple leaders, co-discussion leaders should divide papers among themselves but may not discuss the same paper. Discussion questions should be submitted to the instructor by email

the evening before class. Students should come to the Professor's office hours at least one week before they are scheduled to present to discuss the paper and ensure they understand everything they need to understand about their assigned paper.

Preparation of this presentation should take 4-6 hours. In contrast to the referee report, students should gain a sufficiently thorough understanding of the paper that they can explain it to their classmates effectively.

Assignment 3: Class Demonstration Each student will be responsible for providing a practical demonstration of one week's topic beginning the third week of the term. This will be in the form of a fully functioning Rmarkdown or Jupyter notebook (it should run from start to finish with no errors). The student will lead the class in walking through the notebook demonstrating the code and the output of the notebook. This demonstration can be one of a variety of things:

1. Running a method discussed during the week on an applied problem of interest (e.g. a dataset of interest)
2. Implementing the method "from scratch" and highlighting how it works and specific implementation-specific concerns that must be addressed
3. A simulation detailing the performance of the method (or multiple methods) in a setting where the "truth" is known

Whichever approach the student takes, this should take approximately 20 minutes. The notebook should (perhaps by reducing the number of replicates) be able to be run within a few minutes in-class.

If the class-size demands that multiple students are assigned to a single class, then they will confer with the professor about how to split up the week's session or work together.

Preparation of this demonstration should take 1-2 days of work. While the presentation focuses on understanding the paper, the demonstration ensures that the student has practical understanding of the implementation, how it can be used and/or what its strengths and weaknesses are when applied to data. Students should take a few iterations to ensure that the analysis provides insight into the methods under consideration in the session.

Assignment 4: Final Project The final project is a work of exposition/tutorial in which you explore a topic related to machine learning for causal inference both analytically and computationally. Discuss the foundations for this work in statistics or related fields. Then, choose and focus on some aspect of how this work has developed, competing methodologies or debates with associated benefits and drawbacks, and directions for future research. Demonstrate these ideas with a computational example (i.e. write code). The deliverable is a 5 page paper that should include focused code snippets (if appropriate) and key output as well as a script or notebook and related data that can be run all the way through after setting the working directory. For example, you can write an Rmd or Jupyter notebook where only the most important code and output for understanding your point is shown in the pdf output (there should not be extraneous code in the report). The reader of your report should come away with a good understanding of the strengths and drawbacks of the method/theme as well as how to use them effectively. The class demonstration is a good place to start for this project, but you are not *required* to have your project be an extension of your class demonstration.

You will be graded not only on the accuracy of your analysis, but also on the clarity of your exposition, which includes notation, typesetting (equation alignment, exceeding margins), grammar, figures, tables, etc. All code should be reproducible–please provide any supporting files necessary with your submission. The more difficult it is to read your work, the fewer points

you will receive. Follow a coding style guide (e.g. https://style.tidyverse.org/ and https://peps.python.org/pep-0008/) and stick to it. The expectation is for polished work.

Alternatively, students may submit original work, such as a research proposal for a project they are working on. This option is available with instructor permission only.

Preparation of this report should take up to a week of additional work between extending the analyses of the demonstration and bringing it up to the quality of presentation necessary for the report.

Participation Grade The participation grade assumes that students take part, not as passive consumers of knowledge, but as active participants in the exchange, production, and critique of ideas—their own ideas and the ideas of others. Therefore, students should come to class not only having read and viewed the materials assigned for that day but also prepared to discuss the readings and to contribute thoughtfully to the conversation. This includes reading their colleagues' response papers before class. Participation is marked by its active nature, its consistency, and its quality. To receive the best possible participation grades, students should ask thoughtful questions, build on their colleagues' comments and come prepared to class with specific observations about the readings.

This class is structured as a seminar, so there will be substantial expectations on students to contribute. Students should spend a few hours ahead of class reading the assigned readings carefully. Active reading (i.e. highlighting and taking notes) will ensure that students are prepared to contribute in class.

Important Policies

Late Submission of Assignments For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

4. General Readings

There is no overall textbook for this course, required readings are all papers available online (and which will be posted on GitHub). A few texts may be useful to the student, but they are completely optional. If you're considering getting any of them, come by the Professor's office to browse them first before committing to the sometimes high price.

Optional Textbooks

Causal machine learning. These three are the closest thing the field has to a textbook treatment of the material in this course.

- **Stefan Wager.** *Causal Inference: A Statistical Learning Approach* — *Freely available. The most direct match to this course's perspective: causal inference done with the tools and asymptotics of statistical learning.*
- **Victor Chernozhukov, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis.** *Applied Causal Inference Powered by ML and AI*, 2024 (updated May 2026) — *The other CausalML book. The authors are great, but it doesn't quite make it to the level of "canonical text".*
- **Martin Huber.** *Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R*, MIT Press, 2023 — *One of a very few books focusing on CausalML, and the one with the most attention to applied practice in R.*

Causal inference. General-purpose references for the identification and estimation ideas the course builds on.

- **Miguel A Hernán and James M Robins.** *Causal Inference: What If*, Chapman & Hall/CRC, 2020 — *This comes from an epidemiological perspective, but is a great overview of important causal and statistical concepts.*
- **Guido W Imbens and Donald B Rubin.** *Causal inference in statistics, social, and biomedical sciences.* Cambridge University Press, 2015 — *This is a great general-purpose reference book to have on hand for causal inference.*

Design and econometrics.

- **Alan S Gerber and Donald P Green.** *Field experiments: Design, analysis, and interpretation.* WW Norton, 2012 — *This is still the best general-purpose handbook for experimentation.*
- **Joshua D Angrist and Jörn-Steffen Pischke.** *Mostly harmless econometrics: An empiricist's companion.* Princeton university press, 2009 — *Compellingly written. It's a little dated, now, but still a great resource.*

Session 1: Basics of Design-based Causal Inference / Monte Carlo Simulation 1. September

There is no assigned reading for the first session.

Optional:

- Imbens, G. W., & Rubin, D. B. (2015). Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction, Chapters 1-2. Cambridge University Press.
- Murphy, J. R. (1997). How to read the statistical methods literature: a guide for students. The American Statistician 51(2), 155-157.
- Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. Statistics in Medicine, 38(11), 2074-2102.

Session 2: Covariate Adjustment: when you model the outcome 8. September

Foundational:

- Freedman, D. A. (2008). On regression adjustments to experimental data. Advances in Applied Mathematics, 40(2), 180-193.

Common:

- Bai, Y., Shaikh, A. M., & Tabord-Meehan, M. (forthcoming). "A Primer on the Analysis of Randomized Experiments and a Survey of some Recent Advances." Journal of Political Economy: Microeconomics. arXiv:2405.03910. *Accompanied by the R package* `sreg`*, a natural starting point for this week's demonstration.*

Presenter:

- Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique. Annals of Applied Statistics, 7(1), 295-318.
- Belloni, A., Chernozhukov, V., & Hansen, C. (2014). "Inference on Treatment Effects after Selection among High-Dimensional Controls." Review of Economic Studies, 81(2), 608-650.

- Aronow, P. M., & Middleton, J. A. (2013). A class of unbiased estimators of the average treatment effect in randomized experiments. Journal of Causal Inference, 1(1), 135-154. https://doi.org/10.1515/jci-2012-0009
- Wager, S., Du, W., Taylor, J., & Tibshirani, R. J. (2016). "High-dimensional regression adjustments in randomized experiments." Proceedings of the National Academy of Sciences, 113(45), 12673-12678.

Optional:

- Tsiatis, A. A., Davidian, M., Zhang, M., & Lu, X. (2008). Covariate adjustment for two-sample treatment comparisons in randomized clinical trials: a principled yet flexible approach. Statistics in Medicine, 27(23), 4658-4677.
- Bloniarz, A., Liu, H., Zhang, C.-H., Sekhon, J. S., & Yu, B. (2016). Lasso adjustments of treatment effect estimates in randomized experiments. Proceedings of the National Academy of Sciences, 113(27), 7383-7390.
- Hill, J. L. (2011). Bayesian nonparametric modeling for causal inference. Journal of Computational and Graphical Statistics, 20(1), 217-240.
- Bannick, M. S., Shao, J., Liu, J., Du, Y., Yi, Y., & Ye, T. (2025). "A general form of covariate adjustment in clinical trials under covariate-adaptive randomization." Biometrika, 112(3), asaf029.

Session 3: Balancing Weights: when you model the treatment 15. September

Foundational:

- Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. Biometrika, 70(1), 41-56.

Common:

- Ben-Michael, E., Feller, A., Hirshberg, D. A., & Zubizarreta, J. R. (2021). "The Balancing Act in Causal Inference." arXiv:2110.14831.

Presenter:

- Imai, K., & Ratkovic, M. (2014). "Covariate Balancing Propensity Score." Journal of the Royal Statistical Society: Series B, 76(1), 243-263.
- Arbour, D., Dimmery, D., & Sondhi, A. (2021). Permutation weighting. In International Conference on Machine Learning, 331-341. PMLR.
- Wang, Y., & Zubizarreta, J. R. (2020). "Minimal dispersion approximately balancing weights: asymptotic properties and practical considerations." Biometrika, 107(1), 93-105.

Optional:

- Li, F., Thomas, L. E., & Li, F. (2019). Addressing extreme propensity scores via the overlap weights. American Journal of Epidemiology, 188(1), 250-257.
- Zhao, Q. (2019). "Covariate Balancing Propensity Score by Tailored Loss Functions." Annals of Statistics, 47(2), 965-993.
- Zubizarreta, J. R. (2015). "Stable Weights that Balance Covariates for Estimation With Incomplete Outcome Data." Journal of the American Statistical Association, 110(511), 910-922.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. Political Analysis, 20(1), 25-46.

Session 4: Double Robustness, Double ML and TMLE: when you model both outcome and treatment 22. September

Foundational:

- Bang, H., & Robins, J. M. (2005). "Doubly Robust Estimation in Missing Data and Causal Inference Models." Biometrics, 61(4), 962-973.

Common:

- Kennedy, E. H. (2022). Semiparametric doubly robust targeted double machine learning: a review. arXiv:2203.06469. Parts 1-3.

Presenter:

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. The Econometrics Journal, 21(1), C1-C68. Parts 1-3.
- Schuler, M. S., & Rose, S. (2017). Targeted maximum likelihood estimation for causal inference in observational studies. American Journal of Epidemiology, 185(1), 65-73.
- Bruns-Smith, D., Dukes, O., Feller, A., & Ogburn, E. L. (2025). "Augmented balancing weights as linear regression." Journal of the Royal Statistical Society: Series B, 88(3), 699-723.
- Chernozhukov, V., Newey, W. K., Quintas-Martinez, V., & Syrgkanis, V. (2022). "RieszNet and ForestRiesz: Automatic Debiased Machine Learning with Neural Nets and Random Forests." International Conference on Machine Learning, 3901-3914. PMLR. *Implemented in the* `EconML` *Python package.*

Optional:

- Chernozhukov, V., Newey, W. K., & Singh, R. (2022). "Automatic Debiased Machine Learning of Causal and Structural Effects." Econometrica, 90(3), 967-1027.
- van der Laan, M. J., & Rose, S. (2011). "Targeted Learning: Causal Inference for Observational and Experimental Data", Chapters 4-6. Springer.
- Kang, J. D., & Schafer, J. L. (2007). "Demystifying Double Robustness: A Comparison of Alternative Strategies for Estimating a Population Mean from Incomplete Data." Statistical Science, 22(4), 523-539.

Session 5: Heterogeneous Treatment Effects I: Basics 29. September

Foundational:

- Holland, P. W. (1986). Statistics and causal inference. Journal of the American Statistical Association, 81(396), 945-960.

Common:

- Athey, S., & Imbens, G. W. (2019). "Machine Learning Methods That Economists Should Know About." Annual Review of Economics, 11, 685-725.

Presenter:

- Wager, S., & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. Journal of the American Statistical Association, 113(523), 1228-1242.
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. Proceedings of the National Academy of Sciences, 116(10), 4156-4165.

- Kennedy, E. H. (2020). Towards optimal doubly robust estimation of heterogeneous causal effects. arXiv:2004.14497.
- Chernozhukov, V., Demirer, M., Duflo, E., & Fernández-Val, I. (2025). "Fisher-Schultz Lecture: Generic Machine Learning Inference on Heterogeneous Treatment Effects in Randomized Experiments." Econometrica, 93(4), 1121-1164.

Optional:

- Athey, S., & Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. Proceedings of the National Academy of Sciences, 113(27), 7353-7360.
- Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forests. Annals of Statistics, 47(2), 1148-1178.
- Nie, X., & Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. Biometrika, 108(2), 299-319.

Session 6: Heterogeneous Treatment Effects II: Neural networks 6. October

Foundational:

- Shalit, U., Johansson, F. D., & Sontag, D. (2017). "Estimating Individual Treatment Effect: Generalization Bounds and Algorithms." International Conference on Machine Learning, 3076-3085.

Common:

- Koch, B., Sainburg, T., Geraldo Bastías, P., Jiang, S., Sun, Y., & Foster, J. G. (2025). "A Primer on Deep Learning for Causal Inference." Sociological Methods & Research, 54(2), 397-447.

Presenter:

- Curth, A., & van der Schaar, M. (2021). "On Inductive Biases for Heterogeneous Treatment Effect Estimation." Advances in Neural Information Processing Systems, 34.
- Shi, C., Blei, D., & Veitch, V. (2019). "Adapting Neural Networks for the Estimation of Treatment Effects." Advances in Neural Information Processing Systems, 32.
- Ma, Y., Frauen, D., Javurek, E., & Feuerriegel, S. (2026). "Foundation Models for Causal Inference via Prior-Data Fitted Networks." International Conference on Learning Representations.
- Melnychuk, V., Balazadeh, V., Feuerriegel, S., & Krishnan, R. G. (2026). "Frequentist Consistency of Prior-Data Fitted Networks for Causal Inference." International Conference on Machine Learning. arXiv:2603.12037.

Optional:

- Alaa, A. M., & van der Schaar, M. (2017). "Bayesian Inference of Individualized Treatment Effects using Multi-task Gaussian Processes." Advances in Neural Information Processing Systems, 30.
- Yoon, J., Jordon, J., & van der Schaar, M. (2018). GANITE: Estimation of individualized treatment effects using generative adversarial nets. International Conference on Learning Representations.
- Louizos, C., Shalit, U., Mooij, J. M., Sontag, D., Zemel, R., & Welling, M. (2017). Causal effect inference with deep latent-variable models. Advances in Neural Information Processing Systems, 30.

Session 7: Off-policy evaluation and optimization: Do you need causal effects? 13. October

Foundational:

- Dudík, M., Langford, J., & Li, L. (2011). "Doubly Robust Policy Evaluation and Learning." International Conference on Machine Learning, 1097-1104.

Common:

- Dudík, M., Erhan, D., Langford, J., & Li, L. (2014). "Doubly Robust Policy Evaluation and Optimization." Statistical Science, 29(4), 485-511.

Presenter:

- Athey, S., & Wager, S. (2021). Policy learning with observational data. Econometrica, 89(1), 133-161.
- Fernández-Loría, C., & Provost, F. (2022). "Causal Decision Making and Causal Effect Estimation Are Not the Same…and Why It Matters." INFORMS Journal on Data Science, 1(1), 4-16.
- Su, Y., Dimakopoulou, M., Krishnamurthy, A., & Dudík, M. (2020). "Doubly Robust Off-Policy Evaluation with Shrinkage." Proceedings of Machine Learning Research, 119, 9167-9176.

Optional:

- Uehara, M., Shi, C., & Kallus, N. (2022). "A Review of Off-Policy Evaluation in Reinforcement Learning." arXiv:2212.06355.
- Kern, C., Fischer-Abaigar, U., Schweisthal, J., et al. (2025). Algorithms for reliable decision-making need causal reasoning. Nature Computational Science, 5, 356-360. `https://doi.org/10.1038/s43588-025-00814-9`
- Kitagawa, T., & Tetenov, A. (2018). Who should be treated? Empirical welfare maximization methods for treatment choice. Econometrica, 86(2), 591-616.
- Jiang, N., & Li, L. (2016). Doubly robust off-policy value evaluation. International Conference on Machine Learning, 652-661.
- Kallus, N. (2018). "Balanced Policy Evaluation and Learning." Advances in Neural Information Processing Systems, 31.
- Swaminathan, A., & Joachims, T. (2015). Counterfactual risk minimization: Learning from logged bandit feedback. International Conference on Machine Learning, 814-823.

Session 8: Experimental Design: How should you collect data to make your life easier? 27. October *(no session on 20 October – fall break)*

Foundational:

- Greevy, R., Lu, B., Silber, J. H., & Rosenbaum, P. (2004). Optimal multivariate matching before randomization. Biostatistics, 5(2), 263-275.

Common:

- Higgins, M. J., Sävje, F., & Sekhon, J. S. (2016). "Improving Massive Experiments with Threshold Blocking." Proceedings of the National Academy of Sciences, 113(27), 7369-7376. *Accompanied by the R package* `quickblock`*, a natural starting point for this week's demonstration.*

Presenter:

- Kallus, N. (2018). Optimal a priori balance in the design of controlled experiments. Journal of the Royal Statistical Society: Series B, 80(1), 85-112.

- Harshaw, C., Sävje, F., Spielman, D., & Zhang, P. (2024). "Balancing Covariates in Randomized Experiments with the Gram-Schmidt Walk Design." Journal of the American Statistical Association, 119(548), 2934-2946.
- Morgan, K. L., & Rubin, D. B. (2012). Rerandomization to improve covariate balance in experiments. Annals of Statistics, 40(2), 1263-1282.

Optional:

- Athey, S., & Imbens, G. W. (2017). "The Econometrics of Randomized Experiments." Handbook of Field Experiments, 1, 73-140. arXiv:1607.00698.
- Ribeiro Junior, A. C. H., & Branson, Z. (2025). "Does Rerandomization Help Beyond Covariate Adjustment? A Review and Guide for Theory and Practice." arXiv:2512.05290.
- Zhao, J. (2024). "Experimental Design for Causal Inference Through an Optimization Lens." INFORMS TutORials in Operations Research, 146-188. https://doi.org/10.1287/educ.2024.0277
- Arbour, D., Dimmery, D., Mai, T., & Rao, A. (2022). "Online Balanced Experimental Design." Proceedings of the 39th International Conference on Machine Learning (ICML), 844-864.
- Arbour, D., Dimmery, D., & Rao, A. (2021). Efficient balanced treatment assignments for experimentation. Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS), 3070-3078.

Session 9: Panel data and the new Diff-in-diff: Using time for identification 3. November

Foundational:

- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. Journal of the American Statistical Association, 105(490), 493-505.

Common:

- Roth, J., Sant'Anna, P. H. C., Bilinski, A., & Poe, J. (2023). "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." Journal of Econometrics, 235(2), 2218-2244.

Presenter:

- Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. Journal of Econometrics, 225(2), 200-230.
- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). "Synthetic Difference-in-Differences." American Economic Review, 111(12), 4088-4118.

Optional:

- De Chaisemartin, C., & d'Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. American Economic Review, 110(9), 2964-2996.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. Journal of Econometrics, 225(2), 254-277.
- Ben-Michael, E., Feller, A., & Rothstein, J. (2021). "The Augmented Synthetic Control Method." Journal of the American Statistical Association, 116(536), 1789-1803.
- Sun, L., & Abraham, S. (2021). "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." Journal of Econometrics, 225(2), 175-199.
- Ben-Michael, E., Arbour, D., Feller, A., Franks, A., & Raphael, S. (2023). "Estimating the Effects of a California Gun Control Program with Multitask Gaussian Processes." Annals

of Applied Statistics, 17(2), 985-1016.

Session 10: Partial Identification: when you don't have the ideal data for the task 10. November

Foundational:

- Manski, C. F. (1990). Nonparametric bounds on treatment effects. American Economic Review, 80(2), 319-323.

Common:

- Tamer, E. (2010). "Partial Identification in Econometrics." Annual Review of Economics, 2(1), 167-195.

Presenter:

- Lee, D. S. (2009). "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects." Review of Economic Studies, 76(3), 1071-1102.
- Yadlowsky, S., Namkoong, H., Basu, S., Duchi, J., & Tian, L. (2022). "Bounds on the Conditional and Average Treatment Effect with Unobserved Confounding Factors." Annals of Statistics, 50(5), 2587-2615.

Optional:

- Samii, C., Wang, Y., & Zhou, J. (2026). "Generalizing Trimming Bounds for Endogenously Missing Outcome Data Using Random Forests." Political Analysis, 34(1), 23-37.
- Khan, S., Saveski, M., & Ugander, J. (2024). "Off-Policy Evaluation Beyond Overlap: Partial Identification Through Smoothness." Proceedings of the 41st International Conference on Machine Learning (ICML).
- Kallus, N., Mao, X., & Uehara, M. (2022). "Causal Inference Under Unmeasured Confounding With Negative Controls: A Minimax Learning Approach." Journal of the American Statistical Association, 117(540), 2071-2086.
- Imbens, G. W., & Manski, C. F. (2004). Confidence intervals for partially identified parameters. Econometrica, 72(6), 1845-1857.

Session 11: Response Adaptation and reinforcement learning: when you can interact rather than merely act 17. November

Foundational:

- Robbins, H. (1952). Some aspects of the sequential design of experiments. Bulletin of the American Mathematical Society, 58(5), 527-535.

Common:

- Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction, Chapters 1-3, 6, and 13. MIT Press.

Presenter:

- Murphy, S. A. (2003). "Optimal Dynamic Treatment Regimes." Journal of the Royal Statistical Society: Series B, 65(2), 331-355.
- Russo, D., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018). "A Tutorial on Thompson Sampling." Foundations and Trends in Machine Learning, 11(1), 1-96. Sections 1-3.
- Mnih, V., et al. (2016). "Asynchronous Methods for Deep Reinforcement Learning." Proceedings of Machine Learning Research, 48, 1928-1937.

Optional:

- Luedtke, A. R., & van der Laan, M. J. (2016). Super-learning of an optimal dynamic treatment rule. International Journal of Biostatistics, 12(1), 305-332.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). "Proximal Policy Optimization Algorithms." arXiv:1707.06347.
- Haarnoja, T., Zhou, A., Abbeel, P., & Levine, S. (2018). "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor." International Conference on Machine Learning.
- Ouyang, L., Wu, J., Jiang, X., et al. (2022). "Training Language Models to Follow Instructions with Human Feedback." Advances in Neural Information Processing Systems, 35, 27730-27744.
- Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., & Athey, S. (2021). "Confidence Intervals for Policy Evaluation in Adaptive Experiments." Proceedings of the National Academy of Sciences, 118(15).
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Session 12: Interference: What if units are not independent from one another? 24. November

Foundational:

- Hudgens, M. G., & Halloran, M. E. (2008). Toward causal inference with interference. Journal of the American Statistical Association, 103(482), 832-842.

Common:

- Halloran, M. E., & Hudgens, M. G. (2016). "Dependent Happenings: A Recent Methodological Review." Current Epidemiology Reports, 3(4), 297-305.

Presenter:

- Eckles, D., Karrer, B., & Ugander, J. (2017). "Design and Analysis of Experiments in Networks: Reducing Bias from Interference." Journal of Causal Inference, 5(1), 20150021.
- Aronow, P. M., & Samii, C. (2017). "Estimating Average Causal Effects under General Interference, with Application to a Social Network Experiment." Annals of Applied Statistics, 11(4), 1912-1947.
- Wager, S., & Xu, K. (2021). "Experimenting in Equilibrium." Management Science, 67(11), 6694-6715.

Optional:

- Ugander, J., Karrer, B., Backstrom, L., & Kleinberg, J. (2013). "Graph Cluster Randomization: Network Exposure to Multiple Universes." Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 329-337.
- Athey, S., Eckles, D., & Imbens, G. W. (2018). "Exact p-Values for Network Interference." Journal of the American Statistical Association, 113(521), 230-240.
- Weinstein, B., & Nevo, D. (2026). "Causal inference with misspecified network interference structure." Biometrics, 82(1), ujag023.

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This syllabus is subject to change at the instructor's discretion. All updates will be communicated via Moodle and in class.